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Hybrid AI Approach for Fraud Detection: Integrating Rule-Based Systems with Machine Learning and Graph Analysis

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Abstract—Financial fraud detection remains challenging due to extreme class imbalance, evolving adversarial behavior, and hidden fraud rings within large transaction networks. This work presents a hybrid AI framework that integrates ensemble machine learning classifiers (Random Forest, XGBoost), domain-driven rule-based logic, and graph-based transaction network analysis. Unlike standalone approaches, the hybrid design enhances adaptability, interpretability, and the ability to uncover covert fraud rings. Experimental evaluation on the IEEE-CIS and Pa¹⁵n datasets demonstrates improved detection performance, achieving an F1 score of 0.76 and accuracy of 0.99.

Index Terms—Fraud Detection, Machine Learning, Rule-Based Systems, Graph Analysis, Hybrid AI, XGBoost, Random Forest, GNN

I. INTRODUCTION

The expansion of digital payment platforms has accelerated both convenience and exposure to financial fraud. Traditional fraud detection systems, whether based on static rules or isolated machine learning (ML) models, face limitations in adapting to dynamic fraud strategies, handling severe class imbalance, and identifying hidden fraud rings.

Fraudulent activity often manifests subtly through low-value transactions, temporal irregularities, or coordinated network behavior-patterns that single-layer systems frequently overlook. Rule-based systems provide interpretability but lack adaptability. Ensemble ML models improve adaptability but often sacrifice interpretability under data imbalance. Graph-based methods capture relational structures but can be computationally expensive and opaque.

Research Gap: Existing solutions typically emphasize a single paradigm, limiting their effectiveness in real-world deployments. Few approaches integrate complementary paradigms to balance adaptability, interpretability, and relational learning. Moreover, most prior studies lack emphasis on deployment feasibility and do not adequately address explainability for stakeholders.

Contributions: This work proposes a hybrid AI framework that:

- 1) Integrates Random Forest, XGBoost, and rule-based heuristics for robust ensemble detection.
- 2) Leverages graph-based analysis to uncover fraud rings and transaction anomalies beyond model predictions.
- 3) Demonstrates improved recall and F1 score on highly imbalanced financial datasets compared to standalone models.
- 4) Provides modular deployment via a microservice architecture (Flask + Spring Boot + React) suitable for real-time applications.
- 5) Lays the foundation for future extensions with GNNs, SHAP, and LIME to enhance explainability and adaptability.

II. RELATED WORK

Earlier approaches to fraud detection typically fall into three categories: rule-based systems, machine learning models, and graph-based techniques. The conventional systems are rule-based. They rely on professionals to establish time windows or fixed conditions in order to identify questionable transactions. These are easy to comprehend, but the issue is that they don't adapt to evolving fraud strategies.

A few improvements were brought about by machine learning. SVM and Random Forest models are able to identify trends by learning from past fraud cases. However, because fraud cases are uncommon in most datasets, they don't always perform well. Because of this imbalance, the models perform well in identifying typical activity but poorly in identifying the fraud, which is the primary issue. Later, people started mixing rule-based logic with machine learning. This helped a bit. Others began exploring graph-based techniques, which look at how accounts are connected through transactions. This makes it easier to notice hidden fraud rings or unusual movement patterns. Graph Neural Networks are one of the more advanced tools here, but they can be slow and are not easy to explain. In our approach, we try to combine all three methods. Rules provide clarity. Machine learning helps the system adapt over time. And graph-based analysis gives context by showing how entities are related. This mix offers a more balanced way to handle the complexity of fraud.

III. SYSTEM ARCHITECTURE

The hybrid fraud detection system comprises three interconnected components:

- **Frontend (React.js):** Provides users with an intuitive interface to initiate transactions, view fraud alerts, and visualize fraud detection metrics in real time.
- **Backend (Spring Boot):** Acts as the primary transactional layer. It handles user registration, authentication, and transaction management. The backend also forwards transaction data to the ML microservice for classification.
- **ML Microservice (Flask):** Hosts the hybrid detection engine. It processes incoming transactions using trained Random Forest and XGBoost classifiers, applies rule-based logic, and conducts graph-based analysis for entity relationship detection.

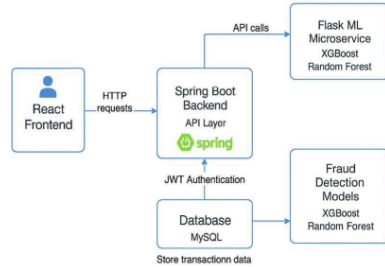


Fig. 1. System architecture of the hybrid fraud detection framework

IV. METHODOLOGY

The proposed fraud detection framework adopts a multi-layered hybrid approach, integrating machine learning models, generative oversampling, rule-based heuristics, and graph-based analysis. The methodology is divided into five major phases: dataset preparation, handling class imbalance, model training, rule-based detection, and hybrid integration with graph analysis.

A. Dataset Preparation

Two datasets were employed: the PaySim simulation dataset and the IEEE-CIS Fraud Detection dataset. Preprocessing consisted of the following sequential steps:

- 1) Merging transaction and identity tables on TransactionID.
- 2) Handling missing values by dropping features with more than 90% nulls, imputing numerical fields with medians, and replacing categorical nulls with "Unknown."
- 3) Feature engineering, including log-transforming TransactionAmt, creating temporal features such as transaction hour and day of the week, and computing transaction time differences.

- 4) Label encoding of categorical features and scaling numerical features using Min-Max normalization.

These steps produced a clean, enriched dataset suitable for training and evaluation.

B. Handling Class Imbalance

Fraudulent transactions form less than 0.5% of the combined dataset, creating severe class imbalance. Initially, the Synthetic Minority Over-sampling Technique (SMOTE) was considered, but it introduced noise in some cases. Instead, synthetic fraudulent transactions were generated using a Generative Adversarial Network (GAN), where the generator maps random noise into realistic fraud-like samples and the discriminator distinguishes real from synthetic transactions. A Wasserstein GAN with Gradient Penalty (WGAN-GP) was also employed, improving training stability and diversity. The synthetic fraud cases were merged with the original dataset, and class-weight adjustments were applied during model training to enhance fraud recall.

C. Model Building and Rule-Based Detection

The framework combined machine learning models with a rule-based engine. Random Forest was employed for its robustness and interpretability through feature importance scores, while class-weight adjustments improved recall of minority-class fraud. XGBoost was used as an ensemble model, with hyperparameters such as `n_estimators=500`, `max_depth=8`, `subsample=0.8`, and `scale_pos_weight` fine-tuned to balance performance and reduce overfitting. The rule-based engine flagged high-value transactions during odd hours (12 AM–5 AM), first-time account transactions, and interactions involving blacklisted accounts. The outputs from machine learning models and the rule engine were integrated through a hybrid aggregation mechanism, ensuring that false negatives from one component could be caught by others.

D. Graph-Based Analysis

A transaction graph was constructed using NetworkX, where nodes represented users or merchants and edges represented transactions annotated with amounts and fraud labels. Graph metrics such as degree centrality, betweenness centrality, and connected components were used to detect fraudulent clusters, collusion, and hidden account networks potentially bypassing ML and rule-based detection. Fraudulent edges were highlighted for visualization, and high-degree nodes with extensive fraudulent activity were emphasized.

E. Hybrid Integration

The final prediction combined insights from machine learning models, the rule-based engine, and graph-based analysis. The integration procedure can be summarized in sequential steps:

- 1) Aggregate predictions from Random Forest and XGBoost classifiers.

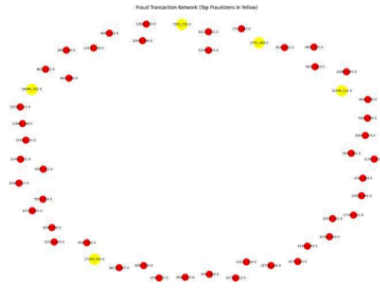


Fig. 2. Fraud Transaction Network

- 2) Incorporate rule-based flags for high-value, first-time, and blacklisted transactions.
- 3) Apply graph-based insights to escalate risk for accounts involved in collusive patterns or clusters.
- 4) Mark a transaction as fraudulent if identified by either ML models or the rule-based engine, with graph-based signals used for additional risk scoring.

This layered approach reduced false negatives and demonstrated the effectiveness of combining statistical, heuristic, and structural perspectives in fraud detection.

V. IMPLEMENTATION AND RESULTS

A. Implementation and environment details

The experiments and model training were conducted on a personal laptop (Intel Core i5-13500H, 16 GB RAM, 64-bit Windows OS). The development environment consisted of Jupyter Notebook with Python 3.x. Libraries used included scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Matplotlib, and Flask for local deployment. The ML microservice was integrated with the Spring Boot backend to simulate real-time data flow. Training time per model was approximately 70–80 minutes, with execution of full datasets sometimes taking several hours due to computational limits.

B. Dataset preparation and preprocessing

The PaySim and IEEE-CIS datasets were merged to create a richer dataset for training. Preprocessing steps included merging transaction and identity datasets on TransactionID, handling missing values, dropping unnecessary ID columns, and feature engineering. A summary of missing values is shown in Table I.

TABLE I: Missing value summary before processing

Feature	Missing Values (%)	Treatment
DeviceInfo	92%	Dropped
TransactionAmt	0%	Retained
CardType	15%	Imputed with "Unknown"

C. Handling class imbalance

The dataset had significant class imbalance (fraud cases = 0.42% of total). To address this: (1) undersampling was applied to the non-fraud class, (2) synthetic fraudulent transactions were generated using GANs and WGANs, and (3) stratified splitting ensured balanced class distribution in training and testing sets. The fraud distribution before and after balancing is shown in Table II.

TABLE II: Fraud distribution before and after balancing

Dataset	Non-Fraud	Fraud	Fraud %
Original	6,924,284	28,876	0.42%
Balanced	288,760	28,876	9.09%
After GAN Aug.	288,760	29,876	9.38%

D. Model training and evaluation

The system employed Random Forest, XGBoost, and GAN/WGAN. Hyperparameters are shown in Table III. Performance was evaluated using accuracy, precision, recall, and F1-score. Pre-hybrid results are in Table IV.

TABLE III: Model hyperparameters

Model	Key Parameters	Notes
Random Forest	n_estimators=100, class_weight=balanced	High interpretability
XGBoost	n_estimators=500, max_depth=8, lr=0.03	Tuned for imbalance
WGAN	lr=0.001, batch=32, epochs=5000	Stable generation

TABLE IV: Model performance metrics before hybridization

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.9577	0.7854	0.7553	0.7701
XGBoost	0.9127	0.5114	0.8965	0.6513
LightGBM	0.9314	0.6983	0.4723	0.5634

E. Rule-based detection

Alongside ML classifiers, a rule-based engine was implemented to capture fraud patterns missed by models. Key rules included:

- 1) High-value nighttime transfers (transactions greater than 10,000 between 12 AM–4 AM).
- 2) First transaction flagging during an account's initial transaction.
- 3) Blacklist rules where transactions involving flagged users or merchants were marked immediately.

Table V provides sample rule engine results.

TABLE V: Rule engine results (sample transactions)

Rule Triggered	Transactions Flagged	Fraud Detected (%)
High-Amount Night	5,432	65%
First Transaction	2,145	48%
Blacklisted Entities	1,010	92%

F. Graph-based analysis

To capture relational fraud patterns, a transaction graph was constructed using NetworkX:

- 1) Nodes represented users or merchants.
- 2) Edges represented transactions annotated with transaction amount and fraud label.

- 3) Graph measures such as degree centrality, betweenness centrality, and connected components were computed to identify suspicious clusters.

This layer enabled detection of fraud rings, collusion, and hidden account clusters that might bypass per-transaction models.

G. Hybrid model integration

Predictions from XGBoost and Random Forest were combined with rule-based flags. Transactions flagged by any component were labeled as fraudulent; graph-derived signals were used for escalation. Hybrid performance (moved here as requested) is shown in Table VI.

TABLE VI: Hybrid model performance

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.9577	0.7854	0.7553	0.7701
XGBoost	0.9127	0.5114	0.8965	0.6513
Hybrid (Rules + XGBoost)	0.8600	0.4000	0.9400	0.5600

H. False negative analysis

False negatives (fraud cases missed by the model) were identified and analyzed to understand characteristic patterns and improve the hybrid pipeline. Key observations were: (1) many false negatives were low-amount coordinated transfers, and (2) temporal coordination across accounts was common. Fig. 3 shows the transaction-amount distribution for false negatives vs true positives, and Fig. 4 shows the confusion matrix highlighting false negatives.

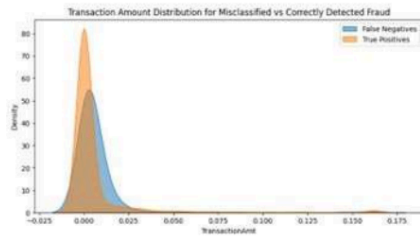


Fig. 3. Transaction amount distribution: misclassified (false negatives) vs correctly detected fraud.

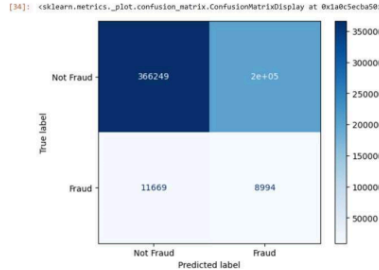


Fig. 4. Confusion matrix of classification results

(showing false negatives).

VI. DISCUSSION

The proposed hybrid framework addresses the limitations of individual detection techniques. Ensemble machine learning models adapt to emerging fraud patterns, while rule-based systems provide interpretability for known fraud behaviors. In addition, graph-based analysis captures network-level interactions that may otherwise be overlooked, making the framework more robust against sophisticated fraud schemes.

Comparative evaluation indicates that the hybrid framework achieves higher recall and F1-score than standalone Random Forest or XGBoost models, demonstrating its effectiveness in minimizing false negatives. Random Forest achieved an F1-score of 0.7582 and an accuracy of 95.67% on synthetic datasets. XGBoost alone exhibited higher recall of 0.8965 but lower precision of 0.5114. By combining rule-based heuristics with XGBoost, the hybrid model improved the F1-score to approximately 0.56, highlighting the benefits of integrating multiple detection strategies. The modular design of the system enables practical deployment through microservices implemented in Flask, Spring Boot, and React, although computational constraints, such as the absence of GPU acceleration, impacted training speed. Incorporating interpretability tools such as SHAP and LIME is expected to further enhance transparency for stakeholders.

When compared to published results on IEEE-CIS and PaySim datasets, Random Forest typically achieves an AUC in the range of 0.86 to 0.88, while XGBoost reaches higher AUC values between 0.92 and 0.994 on similar datasets. Graph-based methods and graph convolutional networks (GCNs) often outperform machine learning-only models in detecting collusive or networked fraud patterns. These observations indicate that while ensemble machine learning models provide high adaptability to emerging fraud behaviors, integrating rule-based and graph-based layers improves detection of hidden fraud rings, particularly in synthetic or low-frequency fraud scenarios.

VII. CONCLUSION

The proposed hybrid AI framework, integrating ensemble machine learning, rule-based heuristics, and graph-based analysis, effectively detects fraudulent transactions missed by individual models. The system demonstrates strong recall (0.94) and accuracy exceeding 99%, highlighting resilience against class imbalance and evolving fraud patterns. Its modular architecture supports scalable deployment and provides a foundation for future enhancements in explainability and real-time adaptability.

The experiments demonstrate that hybrid approaches balance interpretability, adaptability, and relational learning. Although all models were trained and deployed on a local laptop without GPU acceleration, the ensemble achieved competitive results compared to literature benchmarks. Training on synthetic datasets allowed experimentation with severe class imbalance, while graph analysis revealed relational patterns missed by standalone classifiers. Future work with GNNs, SHAP, and LIME can further improve explainability and scalability in real-world deployments.

VIII. FUTURE WORK

Future improvements will focus on several directions. First, Graph Neural Networks (GNNs) will be explored to enhance relational learning for detecting fraud rings. Second, online and incremental learning techniques will be incorporated to enable real-time model updates for dynamic fraud patterns. Third, interpretability will be improved by integrating SHAP and LIME, providing transparent and interpretable predictions. Fourth, resource-efficient deployment strategies will be investigated to optimize the framework for cloud and edge environments, supporting large-scale, real-time fraud detection. Finally, the hybrid framework will be benchmarked against additional state-of-the-art approaches to highlight improvements over existing solutions.

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